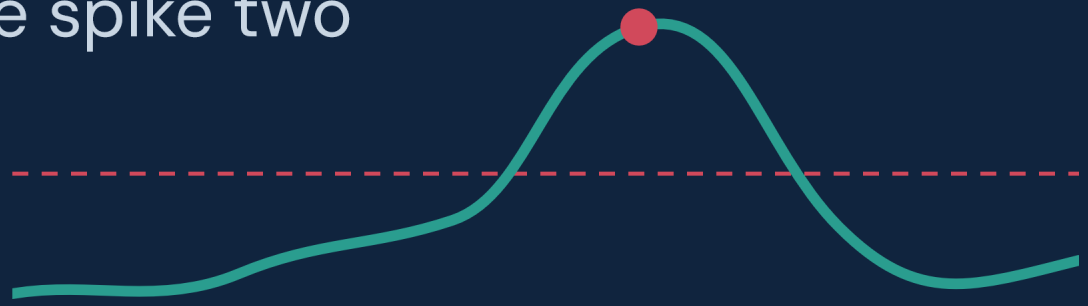
HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE 2026

GlucoTwin

A digital twin that warns of a glucose spike two hours before it happens



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THE PROBLEM

Glucose spikes are noticed after they happen

101 million

people live with diabetes in India, about 11% of the population

Seen too late

A spike usually shows up at the next review or when symptoms appear.

A reading is not a forecast

A CGM shows where glucose is now, not where it is heading.

Two hours is time to act

A walk, a lighter meal or a call to the clinic can still change the outcome.

Source: ICMR-INDIAB study, reported 2023 (101 million with diabetes, 11% of the population)

THE IDEA

A digital twin fuses who the patient is with what is happening now

Static record

Age, BMI, HbA1c, labs, medications and complications: who the patient is.

Live stream

Glucose every 15 minutes, meals, insulin and activity: what is happening now.

The twin

Fuses both and updates with every new reading.

2-hour forecast

Risk that glucose exceeds 180 mg/dL, an alert, and the reasons behind it.

THE DATA

Real clinical data, with the wearable layer simulated

Data stream	Source	Real or simulated
Glucose, every 15 minutes	Shanghai T2DM dataset: 100 patients, 109 recordings of 3 to 14 days	Real
Meals and insulin doses	Logged in the same recordings	Real
Clinical record: age, BMI, HbA1c, labs, medications	Summary sheet of the same dataset	Real
Heart rate and steps	Generated from time of day, age, BMI and meals, never from glucose	Simulated

Dataset: Zhao Q. et al., Scientific Data 10, 35 (2023), CC BY 4.0

THE PREDICTION TASK

Will glucose cross 180 mg/dL within the next 2 hours?

89,974
prediction windows, each with 2 hours of history

14.8%
end in a spike above 180 mg/dL

100
patients, never in both training and testing

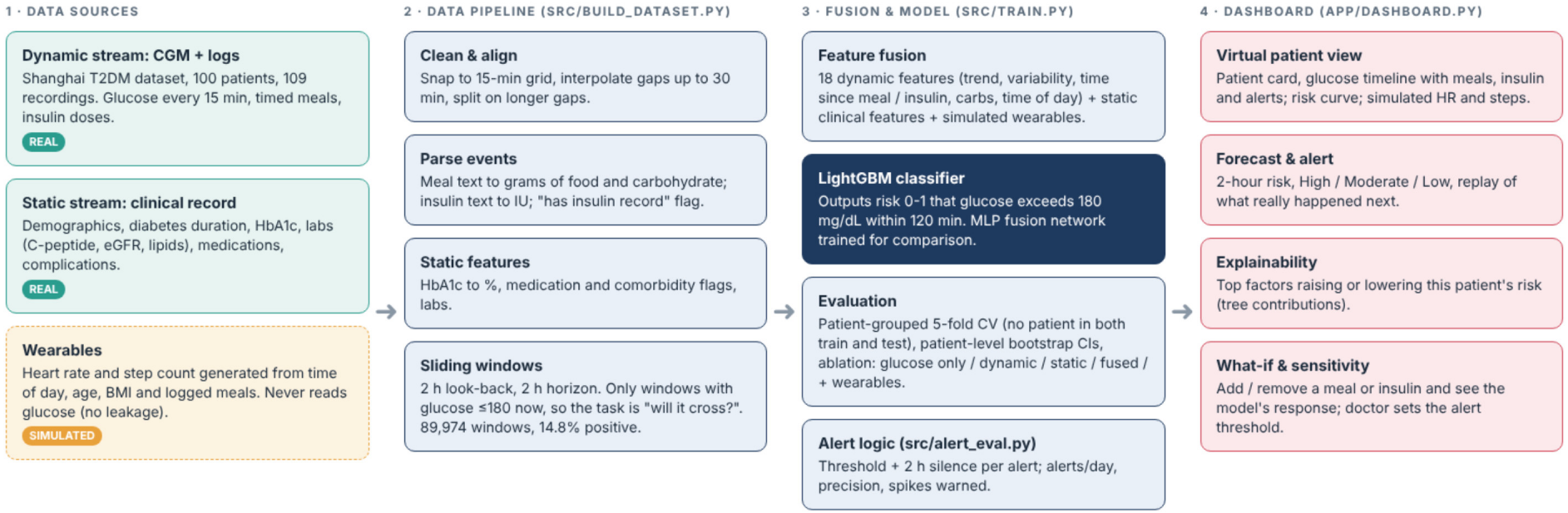
No shortcut. Every window starts at or below 180 mg/dL, so the model must predict a crossing, not repeat a high reading.

Honest scoring. 5-fold cross-validation by patient, with confidence intervals from resampling patients.

GlucoTwin: system architecture

Solid green = real data · dashed orange = simulated

Digital twin that fuses static EHR data with dynamic CGM / wearable streams to forecast a glucose spike (>180 mg/dL) 2 hours ahead in type 2 diabetes



Results (held-out patients)

AUROC 0.73 glucose only → 0.86 with meals, insulin and trend. At threshold 0.5: ~2.5 alerts/day, 63% correct, 78% of spikes warned.

Honest limits

Shanghai cohort, 3-14 days per patient. Heart rate and steps are simulated. Clinical record adds no accuracy once dynamic data is present.

Stack & license

Python, pandas, scikit-learn, LightGBM, Plotly, Streamlit. MIT license. Research prototype, not a medical device.

FEATURES AND MODEL

Three groups of inputs, one gradient boosting model

18

live features

Glucose trend, 2-hour spread, time since meal and insulin, recent carbs, time of day

37

clinical features

Age, sex, BMI, diabetes duration, HbA1c, labs, medications, complications

4

simulated wearable features

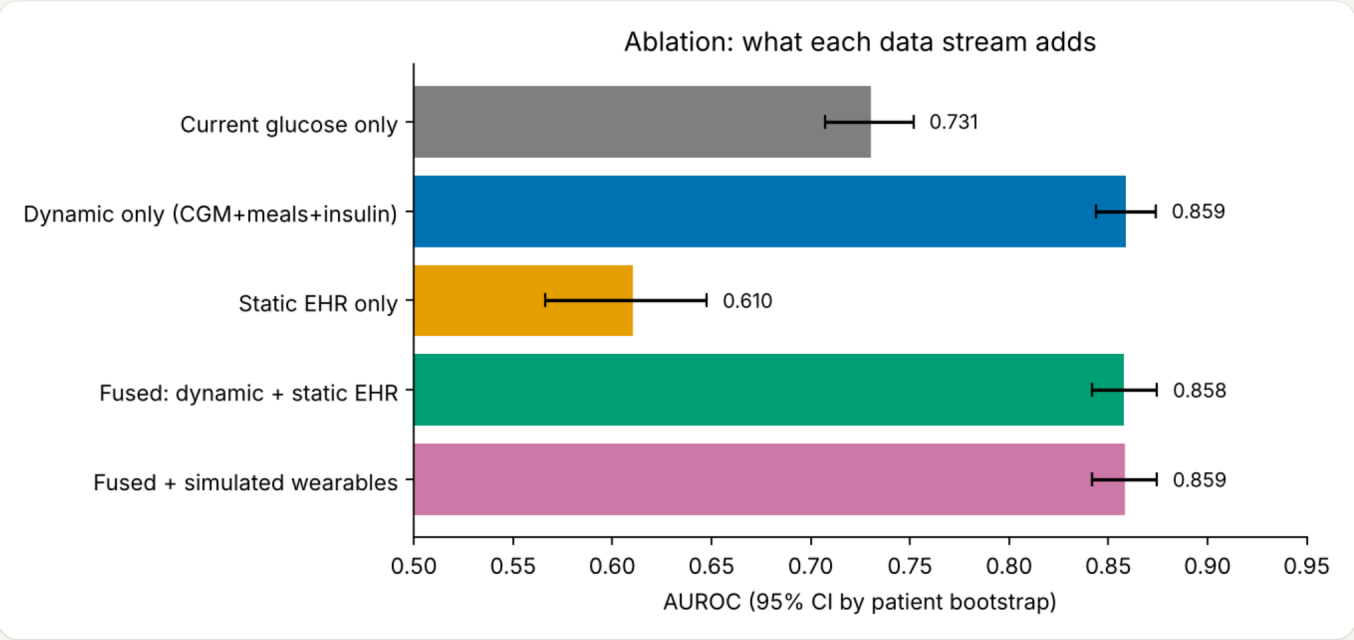
Heart rate and steps over the last 1 to 2 hours

LightGBM classifier

300 shallow trees, no tuning on the test folds. A neural network (MLP) is trained alongside as a comparison.

RESULTS ON HELD-OUT PATIENTS

Meals, insulin and trend lift AUROC from 0.73 to 0.86



0.859

with meals, insulin and glucose trend (95% interval 0.844 to 0.874)

0.731

with current glucose alone (0.707 to 0.752)

5-fold cross-validation by patient, 100 patients, 89,974 windows. Intervals from resampling patients.

WHAT A DOCTOR WOULD SEE

About 2 to 3 alerts a day, set by the doctor

Alert threshold	Alerts per day	Alerts followed by a spike	Spikes warned
0.4	3.1	55%	87%
0.5 (default)	2.5	63%	78%
0.6	1.9	71%	67%
0.7	1.3	79%	53%

Many false alerts are near misses. 60 to 70% of them still reached 160 mg/dL or more.

Held-out patients. After an alert the system stays silent for 2 hours.

AN HONEST FINDING

What fusion did and did not add

Clinical record alone

0.61

AUROC: some signal, far below the live stream (0.86)

Record added to live data

0.859 to 0.858

No gain once meals, insulin and trend are known

Simulated wearables added

0.859 to 0.859

No gain, as designed: the simulator never sees glucose

Neural network with the record

0.84 to 0.74

It falls: with 100 patients it memorizes who they are

With 100 patients, the live stream carries the signal that generalizes.

AUROC on held-out patients. Neural network: MLP, same cross-validation.



THE DOCTOR DASHBOARD

The doctor sees the virtual patient

Patient card: age, HbA1c, medication and conditions.

Timeline: glucose, meals, insulin and the alerts that fired.

Risk curve: the 2-hour forecast with the alert threshold.

Replay: move "now" through the day and see what really happened next.

Heart rate and steps in the dashboard are simulated.

EXPLAIN AND EXPLORE

Add a meal and watch the twin respond



At 05:45 glucose is only 158 mg/dL, yet the twin already flags risk. Adding a 60 g meal 30 minutes earlier raises it by 10 points.

The twin responds to **when** a meal was eaten, much less to its size. This is learned behaviour, not proven cause or treatment advice.

What-if baseline comes from the final model, so it can differ slightly from the held-out curve.

LIMITS

What this prototype does not claim

Small cohort
100 patients, 3 to 14 days each.

Shanghai, not India
Indian validation is future work.

Simulated wearables
No wearable benefit is claimed.

Time of day matters most
It stands in for meal habits, not a medical cause.

Retrospective only
Not clinically validated, not a medical device.

Some alerts are false
63% are right at the default threshold.

WHERE THIS GOES NEXT

From prototype to something clinicians can trust

1

Indian cohort

Validate with consent-based Indian data, handled under the DPDP Act.

2

Real wearables

Replace simulated heart rate and steps with real device data.

3

Sequence models

Try recurrent or temporal networks once there are more patients.

4

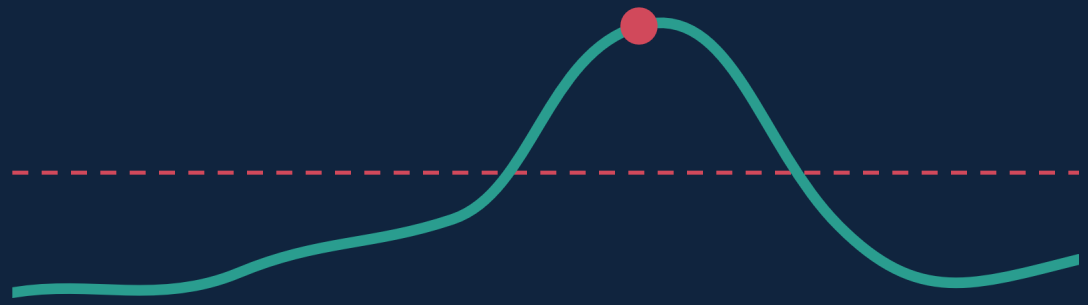
Clinical test

Tune thresholds per patient and test whether alerts help clinicians.

GLUCOTWIN

Thank you

Fusing a patient's record with live data to warn two hours ahead



Code: https://github.com/msahalkb/INDRA_CUSAT

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